MRPL — Equipment Manual

PUMP P-101: CRUDE OIL FEED PUMP

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CHAPTER 1 — INTRODUCTION AND SAFETY

1.1 Scope

This manual covers installation, operation, maintenance, and troubleshooting of Centrifugal Pump P-101 installed at MRPL Process Area 3, Unit 3. The pump is a single-stage, end-suction centrifugal type handling crude oil service at operating temperatures up to 90°C and pressures up to 10 bar(g).

1.2 Safety Warnings

WARNING: Never operate the pump against a closed discharge valve. Cavitation and bearing failure will result within 2 minutes. Minimum flow protection must be maintained at all times. Minimum continuous flow: 80 m^3/h.

WARNING: Do not exceed bearing temperature alarm setpoint of 85°C. If bearing temperature exceeds 85°C, initiate controlled shutdown per SOP-PM-007 Section 4.3 and notify the maintenance team immediately.

CAUTION: Seal flush supply must be maintained at all times when pump is rotating. Loss of seal flush for more than 30 seconds will result in mechanical seal dry-running and premature failure.

CHAPTER 2 — TECHNICAL SPECIFICATIONS

2.1 Performance Data

Design point: 320 m^3/h at 85 m head. Best efficiency point (BEP): 340 m^3/h at 83 m. Specific speed Ns = 1850 rpm(m^3/s). NPSH required at design flow: 4.2 m. Pump efficiency at BEP: 79.5%.

2.2 Mechanical Seal Specification

Seal type: Single mechanical seal, balanced. Shaft diameter: 50 mm. Seal face materials: Silicon carbide (rotating) / Silicon carbide (stationary). Secondary seal: Viton O-rings. Flush arrangement: API Plan 11 (discharge recirculation). Flush flow rate: 3–5 L/min. Flush pressure: Suction + 1.5 bar minimum.

2.3 Bearing Specification

Drive End (DE): SKF 6316/C3, single row deep groove ball bearing. Non-Drive End (NDE): SKF 6313/C3, single row deep groove ball bearing. Lubrication: Grease — Shell Gadus S3 V220C. Regreasing interval: 2000 operating hours or 6 months, whichever is earlier. Grease quantity per regreasing: DE 25g, NDE 15g. Maximum bearing temperature: 85°C (alarm), 95°C (trip).

CHAPTER 3 — OPERATION

3.1 Starting Procedure

Step 1: Confirm all isolation valves open. Step 2: Prime pump casing — open vent valve V-101A until clear liquid flows without air bubbles. Step 3: Confirm seal flush flow (rotameter FR-101 reading 3–5 L/min). Step 4: Close vent valve. Step 5: Start motor via DCS or local panel. Step 6: Slowly open discharge control valve FCV-101 to bring flow to operating point. Step 7: Verify discharge pressure PI-101 reads 9.5–10.0 bar(g). Step 8: Check bearing temperatures after 30 minutes — must be below 75°C at steady state.

3.2 Normal Shutdown

Step 1: Ramp down flow gradually using FCV-101. Step 2: When flow reaches minimum (80 m^3/h), stop motor. Step 3: Close suction isolation valve after motor stops. Step 4: Close discharge valve. Step 5: Seal flush may remain running for 10 minutes after shutdown to cool seal faces.

3.3 Emergency Shutdown

Activate ESD button PB-101-ESD or DCS ESD command. The pump motor will trip and the discharge valve will close automatically. Do NOT restart without maintenance inspection if shutdown was caused by bearing temperature trip or mechanical seal failure alarm.

CHAPTER 4 — TROUBLESHOOTING

Symptom	Probable Cause	Corrective Action
Bearing temperature > 85°C	Insufficient lubrication; bearing failure; misalignment; overload	Check grease; measure vibration; check alignment; reduce flow
High vibration (>4.5 mm/s RMS)	Bearing wear; cavitation; imbalance; misalignment	Check NPSH; inspect bearings; re-balance impeller; realign
Mechanical seal leaking	Seal face wear; dry-running; spring failure; contamination	Inspect seal faces; verify flush supply; replace seal
Low flow / low head	Worn impeller; wrong rotation; air entrainment; blocked suction	Inspect impeller; check rotation direction; prime pump
Noisy operation / cavitation	Low suction head; high fluid temperature; cavitation	Check NPSH available; reduce flow; inspect suction line
Motor overloading	Fluid density too high; overspeeding; mechanical seizure	Check fluid SG; measure power; inspect rotating parts

CHAPTER 5 — MAINTENANCE SCHEDULE

Task	Frequency	Reference
Visual inspection — leaks, vibration, noise	Daily	SOP-PM-007 §2
Bearing temperature check	Daily (DCS)	SOP-PM-007 §2
Seal flush flow check	Weekly	SOP-PM-007 §2
Vibration measurement (handheld)	Monthly	SOP-PM-007 §3
Bearing regreasing	6-monthly or 2000h	SOP-PM-007 §5
Mechanical seal inspection	Annually or on alarm	EM-P101-2019 §7
Full overhaul (bearings, seal, impeller clearance)	3-yearly or on alarm	EM-P101-2019 §7
Pressure test and hydrostatic check	3-yearly	EM-P101-2019 §8